

Project Design Phase Solution Architecture

Date	25 JULY 2026
Team ID	
Project Name	CAB BOOKING
Maximum Marks	4 Marks

Solution Architecture:

Introduction:

The Cab Booking System – MERN Stack Web Application follows a three-tier architecture consisting of the Presentation Layer (Frontend), Application Layer (Backend), and Database Layer. This architecture ensures scalability, maintainability, security, and efficient communication between users, drivers, administrators, and the database.

The system enables users to register, log in securely, search available cabs, book rides, track drivers in real time, make online payments, and manage booking history. Drivers can manage ride requests, while administrators can monitor and manage the entire system through a dedicated dashboard.

Objectives of the Solution Architecture

- Provide a secure and scalable web application.
- Enable seamless communication between frontend, backend, and database.
- Support role-based authentication for Users, Drivers, and Administrators.
- Ensure secure data storage and API communication.
- Simplify maintenance and future enhancements.

Architecture Components

1. Presentation Layer (Frontend)

Technology: React.js, Tailwind CSS, Axios

Functions:

- User Registration/Login
 - Driver Registration/Login
 - Search Available Cabs
 - Cab Booking
 - Fare Estimation
 - Ride Tracking
 - Online Payment Interface
 - Booking History
 - Admin Dashboard
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2. Application Layer (Backend)

Technology: Node.js, Express.js

Functions:

- REST API Development
 - JWT Authentication
 - Password Encryption (bcryptjs)
 - Booking Management
 - Driver Management
 - Payment Processing
 - Ride Status Updates
 - Business Logic
 - Error Handling
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3. Database Layer

Technology: MongoDB with Mongoose

Collections:

- Users
- Drivers
- Cars
- Bookings
- Payments

Stores:

- User Information
- Driver Details
- Vehicle Information
- Ride Details
- Payment Records

SOLUTION ARCHITECHTURE DIAGRAM :

